

Respiratory Outbreak Simulator

1. Project Title

Respiratory Outbreak Simulator: Modelling the Impact of Public Health Interventions

2. Research Question

How do different public health interventions affect the simulated spread of a respiratory infectious disease in a synthetic population?

3. Aim

To develop a computational SIR model in Python to simulate the spread of a respiratory infectious disease and compare the effects of different public health intervention scenarios.

4. Study Population

A synthetic population of **5,000 individuals** will be used.

5. Model

SIR Model

- **S = Susceptible:** people who can become infected
- **I = Infectious:** people who are currently infectious
- **R = Recovered/Removed:** people who are no longer infectious under the model assumptions

6. Initial Population

- Susceptible = **4,990**
- Infectious = **10**
- Recovered/Removed = **0**
- Total = **5,000**

7. Baseline Scenario

The first simulation will represent an outbreak **without any intervention**.

This baseline will later be compared with intervention scenarios such as vaccination and transmission reduction.

8. Data

The project will use **synthetic data** created for modelling purposes and will not use identifiable patient data.

9. Model Parameters

Transmission parameter (β): 0.30 per day

This represents the illustrative rate at which infection spreads through the susceptible population under the model assumptions.

Recovery/removal parameter (γ): 0.20 per day

This represents the illustrative rate at which infectious individuals leave the infectious compartment.

Approximate average infectious period: 5 days

Calculated as:

$$1/\gamma = 1/0.20 = 5 \text{ days}$$

Initial reproduction number (R_0): 1.5

Calculated as: $R_0 = \beta/\gamma = 0.30/0.20 = 1.5$

10. SIR Mathematical Model

The population is divided into three compartments:

S = Susceptible

I = Infectious

R = Recovered/Removed

The model describes how people move between these compartments:

$$S \rightarrow I \rightarrow R$$

The three equations are:

$$\begin{aligned}\frac{dS}{dt} &= -\beta \frac{SI}{N} \\ \frac{dI}{dt} &= \beta \frac{SI}{N} - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

Plain-English explanation**Equation 1 - Susceptible**

$$\frac{dS}{dt} = -\beta \frac{SI}{N}$$

This calculates how many susceptible people are becoming infected.

Equation 2 - Infectious

$$\frac{dI}{dt} = \beta \frac{SI}{N} - \gamma I$$

This calculates how the number of infectious people changes.

It has two parts:

new infections – recoveries

Equation 3 - Recovered

$$\frac{dR}{dt} = \gamma I$$

This calculates how many infectious people move into the recovered/removed group.

Now let's actually use the numbers

This is the important part.

At the beginning of our simulation:

Variable	Value
S	4,990
I	10
R	0
N	5,000
β	0.30
γ	0.20

Now calculate new infections:

$$\beta \frac{SI}{N}$$

Put in our numbers:

$$0.30 \times \frac{4990 \times 10}{5000}$$

This gives approximately:

$$2.994$$

So the model says approximately 3 new infections per day at this particular starting point.

Now calculate recoveries:

$$\gamma I$$

$$0.20 \times 10 = 2$$

So approximately 2 infectious people recover/leave the infectious compartment per day at this starting point.

Therefore:

Susceptible

$$4,990 - 2.994 = 4,987.006$$

Infectious

$$10 + 2.994 - 2 = 10.994$$

Recovered

$$0 + 2 = 2$$

So after approximately one day, our simplified calculation gives:

State	Starting	After ~1 day
Susceptible	4,990	4,987.006
Infectious	10	10.994
Recovered	0	2
Total	5,000	5,000

This is what the computer will eventually do

It will repeat this process:

Calculate new infections → calculate recoveries → update S/I/R → repeat
for day 2, day 3, day 4, and so on.

11. Simulation Period

The SIR model will simulate the respiratory disease outbreak over a period of 180 days.

The model will calculate the number of susceptible, infectious and recovered/removed individuals at successive time points throughout the simulation.

Then Point 12 - Simulation Scenarios

We need something to compare.

We'll eventually run the model four times:

Scenario 1 - Baseline

No intervention.

5,000 people



Disease spreads



Observe outbreak

Scenario 2 - Vaccination

We will introduce an illustrative vaccination scenario.

Scenario 3 - Transmission reduction

We will model an intervention that reduces transmission, representing measures such as masking or reduced contact.

Scenario 4 - Combined intervention

Vaccination + transmission reduction.

12. Simulation Scenarios

Four scenarios will be simulated:

- 1. Baseline: No public-health intervention.**
- 2. Vaccination: An illustrative vaccination intervention.**
- 3. Transmission reduction: An intervention that reduces the effective transmission rate.**
- 4. Combined intervention: Vaccination combined with transmission reduction.**

The baseline scenario will provide the reference point against which intervention scenarios will be compared.

13. Model Outcomes

The model will be used to generate and compare the following outcomes:

- 1. Peak infectious population – the maximum number of infectious individuals at the same time.**
- 2. Day of epidemic peak – the day on which the infectious population reaches its maximum.**
- 3. Cumulative infections – the total number of individuals who have moved through the infection process during the simulation.**
- 4. Final susceptible population – the number of people remaining susceptible at the end of the simulation.**

5. Epidemic curves – graphical representation of changes in susceptible, infectious and recovered/removed populations over time.
6. Scenario comparison – comparison of outcomes between the baseline and intervention scenarios.

14. Baseline Simulation Results

The baseline SIR simulation was run for 180 days using the illustrative model parameters $\beta = 0.30$ per day and $\gamma = 0.20$ per day.

The simulated epidemic reached a peak infectious population of **330.729** individuals on Day 52.

By Day 180, the model contained approximately:

- 2,042.84 susceptible individuals
- 0.036 infectious individuals
- 2,957.12 recovered/removed individuals

These results represent the output of the mathematical model under the specified assumptions and should not be interpreted as a forecast of a real-world outbreak.

15. Vaccination Scenario

A vaccination scenario will be simulated using an illustrative baseline vaccination coverage of 40% and an assumed vaccine effectiveness of 80%.

The effective protected population will therefore be calculated as:

$$5000 \times 0.40 \times 0.80 = 1600$$

The vaccination scenario will then be compared with the baseline scenario using peak infectious population, timing of the epidemic peak and cumulative simulated infections.

16. Baseline vs Vaccination Results

The baseline simulation produced a peak infectious population of **330.729** individuals on Day 52.

The vaccination scenario produced a peak infectious population of **10.494** individuals on Day 28.

The simulated peak infectious population was therefore approximately **96.8% lower** under the vaccination scenario compared with the baseline.

These findings represent a comparison under the specified model assumptions and should not be interpreted as real-world estimates of vaccine effectiveness.

17. Transmission Reduction Scenario

A transmission-reduction intervention will be introduced from Day 30 of the simulation.

The intervention will reduce the effective transmission parameter by 50%, giving:

$$\beta_{\text{intervention}} = 0.30 \times 0.50 = 0.15$$

The scenario will be compared with the baseline and vaccination scenarios using the same model outcomes.

This intervention effect is an illustrative modelling assumption and is not intended to represent the effectiveness of a specific real-world intervention.

18. Transmission Reduction Simulation Results

The transmission-reduction scenario introduced a 50% reduction in the effective transmission parameter from Day 30.

The simulation produced a peak infectious population of 129.822 individuals on Day 29, compared with 330.729 individuals on Day 52 in the baseline scenario.

Under the model assumptions, the transmission-reduction scenario therefore produced an approximately 60.7% lower simulated peak infectious population than the baseline.

The peak occurred immediately before the intervention start date, illustrating the importance of intervention timing in epidemic modelling.

These results are simulation outputs under illustrative assumptions and should not be interpreted as estimates of the effectiveness of a specific real-world intervention.

19. Combined Intervention Scenario

The combined intervention scenario will combine the illustrative vaccination assumption of 40% coverage with 80% effectiveness and a 50% reduction in effective transmission from Day 30.

The combined scenario will be compared with the baseline, vaccination-only and transmission-reduction scenarios using the predefined model outcomes.

20. Combined Intervention Simulation Results

The combined intervention scenario combined the illustrative vaccination assumption with a 50% reduction in effective transmission from Day 30.

The simulation produced a peak infectious population of 10.494 individuals on Day 28, the same peak value and timing observed in the vaccination-only scenario.

Under the model assumptions, the transmission-reduction component introduced on Day 30 did not further reduce the peak because the vaccination scenario had already reached its maximum infectious population before the transmission intervention began.

This illustrates the importance of intervention timing when using computational epidemic models.

21. Final Comparative Results

The four simulated scenarios were compared using peak infectious population, timing of the epidemic peak and cumulative simulated infections.

Scenario	Peak infectious population	Peak day	Cumulative simulated infections
Baseline	330.729	Day 52	2,957.124
Vaccination	10.494	Day 28	279.811
Transmission reduction	129.822	Day 29	665.939
Combined intervention	10.494	Day 28	80.822

21. Final Comparative Results

The four intervention scenarios were compared using peak infectious population, timing of the epidemic peak and cumulative simulated infections.

Scenario	Peak infectious population	Peak day	Cumulative simulated infections
Baseline	330.729	Day 52	2,957.124
Vaccination	10.494	Day 28	279.811
Transmission Reduction	129.822	Day 29	665.939
Combined Intervention	10.494	Day 28	80.822

Under the specified illustrative assumptions, all intervention scenarios produced a lower simulated infectious peak than the baseline scenario. The vaccination and combined scenarios produced the lowest simulated peak, while the combined scenario produced the lowest cumulative simulated infections.

The combined scenario did not further reduce the peak compared with vaccination alone because the transmission-reduction intervention was introduced on Day 30, after the simulated peak on Day 28.

These findings represent model outputs under illustrative assumptions and should not be interpreted as real-world estimates of intervention effectiveness.

22. Sensitivity Analysis

Sensitivity analysis will be conducted to examine how changes in the assumed transmission parameter (β) influence the simulated outbreak.

Three illustrative β values will be assessed:

- $\beta = 0.25$ — lower transmission assumption
- $\beta = 0.30$ — baseline transmission assumption
- $\beta = 0.35$ — higher transmission assumption

The recovery/removal parameter will remain fixed at:

$$\gamma = 0.20$$

For each β value, the model will be re-run and the following outcomes compared:

1. Peak infectious population
2. Day of epidemic peak
3. Cumulative simulated infections

The purpose of the sensitivity analysis is to assess how dependent the model results are on the assumed transmission parameter.

These parameter values are illustrative and are used to demonstrate computational sensitivity analysis rather than to represent estimates for a specific real-world pathogen.

23. Sensitivity Analysis Results

Sensitivity analysis was conducted by varying the transmission parameter β while keeping $\gamma = 0.20$ constant.

β	Peak infectious population	Peak day	Cumulative simulated infections
0.25	117.238	Day 76	1,882.559
0.30	330.729	Day 52	2,957.124
0.35	570.905	Day 39	3,616.966

The results demonstrate substantial sensitivity of the simulated epidemic to the assumed transmission parameter. Increasing β resulted in a larger and earlier simulated epidemic peak and increased cumulative simulated infections.

These findings demonstrate that model outputs are dependent on parameter assumptions and reinforce the importance of parameter estimation, uncertainty assessment and sensitivity analysis in computational epidemiological modelling.

24. Sensitivity Analysis of the Recovery/Removal Parameter

A second sensitivity analysis will examine the effect of varying the recovery/removal parameter (γ) while maintaining the baseline transmission parameter $\beta = 0.30$.

Three illustrative γ values will be assessed:

- $\gamma = 0.15$ — slower recovery/removal
- $\gamma = 0.20$ — baseline value
- $\gamma = 0.25$ — faster recovery/removal

The corresponding approximate infectious periods will be calculated as $1/\gamma$.

The analysis will compare:

1. Peak infectious population
2. Day of epidemic peak
3. Cumulative simulated infections

The purpose is to assess how sensitive the simulated outbreak is to assumptions about the rate at which infectious individuals leave the infectious compartment.

These parameter values are illustrative and are used to demonstrate computational sensitivity analysis rather than to represent estimates for a specific real-world pathogen.

25. Gamma Sensitivity Analysis Results

Sensitivity analysis was conducted by varying the recovery/removal parameter γ while maintaining $\beta = 0.30$.

γ	Approx. infectious period	R_0	Peak infectious population	Peak day	Cumulative simulated infections
0.15	6.67 days	2.0	800.079	Day 42	4,030.702
0.20	5.00 days	1.5	330.729	Day 52	2,957.124
0.25	4.00 days	1.2	83.318	Day 68	1,610.970

The analysis demonstrates that the simulated outbreak is sensitive to the assumed recovery/removal rate. Lower γ values correspond to longer infectious periods and higher R_0 values, resulting in larger and earlier simulated epidemic peaks. Higher γ values correspond to shorter infectious periods, lower R_0 values and smaller simulated peaks.

These results demonstrate the importance of parameter uncertainty in interpreting compartmental epidemic models.

26. Monte Carlo Uncertainty Analysis

A Monte Carlo uncertainty analysis will be conducted to assess how variation in the transmission and recovery/removal parameters affects the simulated epidemic.

Illustrative parameter ranges will be used:

- β : **0.25–0.35**
- γ : **0.15–0.25**

The model will be repeatedly simulated using randomly sampled combinations of β and γ .

For each simulation, the following outcomes will be recorded:

1. β
2. γ
3. R_0
4. Peak infectious population
5. Day of epidemic peak
6. Cumulative simulated infections

The distribution of simulated outcomes will be examined to demonstrate uncertainty in model results.

The parameter ranges are illustrative and are used to demonstrate computational uncertainty analysis rather than to represent empirically estimated ranges for a specific pathogen.

27. Monte Carlo Pilot Results

A preliminary Monte Carlo simulation was conducted using 10 randomly sampled combinations of β and γ within the predefined illustrative ranges.

Across the 10 simulations:

- β varied between approximately **0.253 and 0.339**
- γ varied between approximately **0.151 and 0.218**
- R_0 varied between approximately **1.30 and 2.20**
- Peak infectious population varied from **157.03 to 976.84**
- The simulated epidemic peak occurred between **Day 37 and Day 66**

The wide range of simulated outcomes demonstrates that uncertainty in transmission and recovery/removal parameters can substantially influence epidemic model outputs.

This 10-run analysis is a **pilot demonstration of the Monte Carlo approach**. A larger number of simulations will be used for the final uncertainty analysis.

27. Monte Carlo Uncertainty Analysis Results

A Monte Carlo simulation comprising 1,000 model runs was conducted using randomly sampled values of β between 0.25 and 0.35 and γ between 0.15 and 0.25.

The simulated peak infectious population had the following distribution:

Statistic	Peak infectious population
Minimum	11.336
5th percentile	46.476
Median	328.374
95th percentile	838.264
Maximum	1,024.882

The median simulated peak was 328.374, which was close to the deterministic baseline result of 330.729.

However, the wide 5th–95th percentile range demonstrates substantial uncertainty in the simulated epidemic peak when the transmission and recovery/removal parameters are varied.

These results represent model-based uncertainty under the specified illustrative parameter ranges and should not be interpreted as predictions or confidence intervals for a real-world outbreak.

28. Monte Carlo Distribution of Epidemic Peaks

The 1,000-run Monte Carlo analysis produced a right-skewed distribution of simulated epidemic peak sizes.

The median simulated peak was **328.374 infectious individuals**, while the 5th and 95th percentiles were **46.476 and 838.264**, respectively. The maximum simulated peak was **1,024.882**.

The distribution demonstrates substantial model-based uncertainty arising from variation in the transmission and recovery/removal parameters. Most simulated outcomes were concentrated at lower peak values, with a longer upper tail representing scenarios with substantially larger epidemic peaks.

These results reflect uncertainty within the specified model structure and illustrative parameter ranges and should not be interpreted as a probabilistic forecast of a real-world outbreak.

29. Model Assumptions and Limitations

The SIR model provides a simplified framework for exploring respiratory disease transmission and intervention scenarios. The following assumptions and limitations apply:

Assumption / limitation	Implication
Homogeneous mixing	All individuals are assumed to have similar opportunities for contact and transmission; real contact networks are more complex.
Three compartments only	The model does not explicitly represent an exposed or latent period before infectiousness.
Constant population	Births, deaths and migration are not represented during the simulation period.
Fixed model parameters within scenarios	β and γ do not vary continuously with behaviour, seasonality or changing conditions.
Simplified vaccination representation	Effectively protected individuals are placed in the recovered/removed compartment rather than using a separate vaccinated compartment.
Deterministic baseline model	The baseline SIR simulation does not represent random individual transmission events.
No age or demographic structure	Differences in susceptibility, contact patterns and outcomes by age are not modelled.
No explicit healthcare system capacity	Hospitalisation, ICU demand and healthcare-resource constraints are not directly modelled.
Illustrative intervention effects	Intervention effects are assumptions for demonstrating scenario analysis rather than estimates of real-world effectiveness.
No empirical calibration	The model has not been fitted to real surveillance or outbreak data.

Model improvement opportunities

Future development could include:

1. **SEIR modelling** to introduce an exposed/latent compartment.
2. **Age-stratified modelling** to represent different contact patterns.
3. **Separate vaccination compartment** to model vaccination more explicitly.
4. **Stochastic modelling** to represent random transmission.
5. **Healthcare-capacity modelling** to examine hospitalisation and service pressure.
6. **Evidence-based parameter estimation and calibration** using real epidemiological data.
7. **Agent-based modelling** to represent individual-level contact networks.

30. Model Validation and Quality Checks

The computational model will undergo basic validation and quality checks to ensure that the simulation behaves consistently with the mathematical structure of the SIR model.

The following checks will be performed:

1. **Population conservation:**
The sum of susceptible, infectious and recovered/removed compartments should

remain approximately equal to the total population of 5,000 throughout the simulation.

2. **Non-negative compartments:**
Susceptible, infectious and recovered/removed populations should not become negative.
3. **Initial-condition verification:**
The simulation should begin with 4,990 susceptible, 10 infectious and 0 recovered/removed individuals.
4. **Trajectory checks:**
The susceptible compartment should generally decline, the recovered/removed compartment should increase, and the infectious compartment should demonstrate an epidemic rise and subsequent decline under the baseline assumptions.
5. **Reproducibility:**
Stochastic analyses using fixed random seeds should produce reproducible results when the same code and parameter ranges are used.

These checks are intended as computational consistency and quality-control checks rather than external validation against observed epidemiological data.

31. Model Validation Results

The baseline 180-day SIR simulation underwent a series of computational quality checks.

All validation checks passed:

Validation check	Result
Initial conditions correct	Passed
Population conserved at 5,000	Passed
Maximum population error	0.0
Negative compartment values	0
Susceptible population decreased	Passed
Recovered/removed population increased	Passed
Overall validation	Passed

The validation confirms that the implemented baseline simulation preserved the total population, maintained non-negative compartment values and followed the expected direction of change in the susceptible and recovered/removed compartments.

These checks provide evidence of **computational consistency of the implementation**, but do not constitute validation of the model against real-world epidemiological observations.

32. Extension from SIR to SEIR

The baseline SIR model will be extended to an SEIR structure to represent an exposed or latent stage between infection and infectiousness.

The SEIR compartments will be:

- **S — Susceptible:** individuals who can become infected.
- **E — Exposed:** individuals who have been infected but are not yet infectious under the model assumptions.
- **I — Infectious:** individuals who can transmit the infection.
- **R — Recovered/Removed:** individuals who are no longer infectious under the model assumptions.

The model structure will therefore change from:

$$S \rightarrow I \rightarrow R$$

to:

$$S \rightarrow E \rightarrow I \rightarrow R$$

A new parameter, σ (**sigma**), will control the transition from the exposed compartment to the infectious compartment.

The average exposed period will be represented approximately by:

$$1/\sigma$$

The SEIR extension will allow the project to examine how introducing a latent/exposed stage affects the simulated epidemic trajectory.

Parameter values for σ will be selected and justified separately before the SEIR simulation is implemented.

33. Disease Specification for the SEIR Extension

The SEIR extension will model an **illustrative seasonal influenza outbreak**.

The disease specification will allow the model parameters to be linked to an explicitly defined epidemiological context rather than treating the exposed period as an arbitrary value.

The SEIR model will therefore include:

- **S:** Susceptible
- **E:** Exposed/latent
- **I:** Infectious
- **R:** Recovered/Removed

The exposed-to-infectious transition parameter, σ , will be determined after reviewing appropriate epidemiological evidence.

The project will clearly distinguish between:

1. **Evidence-supported epidemiological parameters**
2. **Illustrative modelling assumptions**
3. **Parameters derived mathematically from other assumptions**

No SEIR parameter will be presented as empirical evidence unless an appropriate source supports it.

34. Evidence-Informed SEIR Parameter: Exposed-to-Infectious Transition

For the SEIR extension, the exposed-to-infectious transition parameter σ (sigma) will represent the rate at which individuals move from the exposed compartment to the infectious compartment.

CDC reports that influenza has a typical incubation period of approximately 1–4 days, with an average around 2 days. However, incubation period and latent period are not equivalent because influenza transmission can occur before symptom onset.

A recent household-transmission analysis estimated a mean influenza latent period of approximately 0.9 days in its primary analysis.

For the initial SEIR model:

$$\text{mean latent period} = 0.9 \text{ days}$$

Therefore:

$$\sigma = \frac{1}{0.9} = 1.11 \text{ per day}$$

The project will use $\sigma = 1.11/\text{day}$ as an initial evidence-informed modelling parameter.

This value will be treated as an estimate subject to uncertainty rather than a fixed biological constant. Sensitivity and uncertainty analysis will subsequently examine how changes in σ affect model outcomes.

35. Initial SEIR Model Implementation

The SEIR model was implemented in Python using four compartments: susceptible (S), exposed (E), infectious (I) and recovered/removed (R).

Using the initial conditions:

- $S = 4,990$
- $E = 0$
- $I = 10$
- $R = 0$

and illustrative parameters:

- $\beta = 0.30$
- $\gamma = 0.20$
- $\sigma = 1.11$

the one-day simulation produced:

- $S = 4,987.006$
- $E = 2.994$
- $I = 8.000$
- $R = 2.000$

The result demonstrates the structural difference between SIR and SEIR: newly infected individuals enter the exposed compartment before transitioning into the infectious compartment.

36. SEIR Simulation Results

The evidence-informed SEIR model was simulated for 180 days using:

- $\beta = 0.30$
- $\gamma = 0.20$
- $\sigma = 1.11$

The simulation produced a peak infectious population of **279.054 individuals on Day 64**.

At the end of the 180-day simulation:

- Susceptible = **2,048.235**
- Exposed = **0.030**
- Infectious = **0.251**
- Recovered/Removed = **2,951.485**

Compared with the baseline SIR model, which produced a peak infectious population of **330.729 on Day 52**, the SEIR model produced a lower and later infectious peak under the same β and γ assumptions.

The difference demonstrates how introducing an exposed/latent compartment can alter the simulated epidemic trajectory.

These results are model outputs and should not be interpreted as empirical estimates of a real influenza outbreak.

37. SIR vs SEIR Comparative Analysis

The baseline SIR and extended SEIR models were compared using the simulated infectious population over 180 days.

Model	Peak infectious population	Peak day
SIR	330.729	Day 52
SEIR	279.054	Day 64

The SEIR model produced a peak infectious population approximately **15.62% lower** than the SIR model and shifted the simulated epidemic peak from Day 52 to Day 64.

The difference reflects the introduction of an exposed compartment in the SEIR structure, creating a transition between infection and infectiousness.

These results demonstrate that **model structure is itself an important source of variation in computational epidemic modelling**. The comparison should not be interpreted as evidence that one model is empirically superior without validation against appropriate epidemiological data.

38. SEIR Transmission-Reduction Scenario

The SEIR model will be extended to include a time-dependent transmission-reduction intervention.

The intervention will be introduced from **Day 30** and will reduce the effective transmission parameter by **50%**.

The model parameters will therefore be:

- $\beta = 0.30$ before Day 30
- $\beta = 0.15$ from Day 30 onward
- $\gamma = 0.20$
- $\sigma = 1.11$

The intervention scenario will be compared with the baseline SEIR model using:

1. Peak infectious population
2. Day of epidemic peak
3. Cumulative simulated infections
4. Epidemic trajectory

5. 39. SEIR Intervention Results

6. The baseline SEIR model was compared with a transmission-reduction scenario in which the effective transmission parameter was reduced by 50% from Day 30.

Scenario	Peak infectious population	Peak day
Baseline SEIR	279.054	Day 64
SEIR transmission reduction	72.841	Day 30

7. Under the specified illustrative assumptions, the transmission-reduction scenario produced an approximately **73.9% lower simulated peak infectious population** than the baseline SEIR model.

8. The simulated peak occurred on Day 30, coinciding with the start of the modelled intervention. Following the intervention, the infectious population declined substantially and reached approximately 0.014 individuals by Day 180.
9. These results demonstrate how introducing a time-dependent change in transmission can alter the trajectory of an SEIR model. The effect size is model-dependent and should not be interpreted as an empirical estimate of real-world intervention effectiveness.

40. Cross-Model Comparison

The project will compare the SIR and SEIR model structures under baseline and transmission-reduction scenarios.

The comparison will examine:

1. Peak infectious population
2. Timing of epidemic peak
3. Change in peak under the transmission-reduction scenario
4. Differences in epidemic trajectory between SIR and SEIR
5. Implications of adding an exposed/latent compartment

The current results are:

Model	Scenario	Peak infectious population	Peak day
SIR	Baseline	330.729	Day 52
SIR	Transmission reduction	129.822	Day 29
SEIR	Baseline	279.054	Day 64
SEIR	Transmission reduction	72.841	Day 30

The comparison will be used to demonstrate how both **model structure** and **intervention timing** can influence computational epidemic outcomes.

The analysis will distinguish model-derived findings from empirical epidemiological evidence and will not treat differences between model structures as evidence that one model is inherently more accurate.

41. Final Cross-Model Comparative Results

The final analysis compared the baseline and transmission-reduction scenarios across both SIR and SEIR model structures.

Model	Scenario	Peak infectious population	Peak day	Peak reduction vs model baseline
SIR	Baseline	330.729	Day 52	—
SIR	Transmission Reduction	129.822	Day 29	60.75%

Model	Scenario	Peak infectious population	Peak day	Peak reduction vs model baseline
SEIR	Baseline	279.054	Day 64	—
SEIR	Transmission Reduction	72.841	Day 30	73.90%

The comparison demonstrates that both model structure and intervention assumptions influence simulated epidemic outcomes. The SEIR model produced a lower and later baseline infectious peak than the SIR model, while the simulated transmission-reduction scenario produced different peak reductions across the two model structures.

The observed differences represent model-dependent simulation outputs, rather than evidence that one model or intervention is inherently superior. Empirical validation would be required to determine which model better represents a particular real-world outbreak.

42. Research Report and Reproducibility

The completed project will be consolidated into a reproducible research report documenting the research question, mathematical model structures, parameter assumptions, computational implementation, intervention scenarios, sensitivity analyses, Monte Carlo uncertainty analysis, validation checks, comparative results, limitations and potential future extensions.

The project will be organised so that code, model outputs, figures and documentation can be traced from the research question through to the final findings.

The GitHub repository will preserve the computational workflow and provide sufficient documentation for another researcher to understand and reproduce the analysis.

43. PhD-Relevant Computational Skills Demonstrated

This project demonstrates practical experience in:

- Translating epidemiological concepts into mathematical compartmental models.
- Implementing SIR and SEIR models in Python.
- Managing model parameters including β , γ and σ .
- Running time-based computational simulations.
- Generating and storing reproducible simulation datasets.
- Creating epidemiological visualisations using Python.
- Comparing alternative intervention scenarios.
- Conducting one-way sensitivity analysis.
- Conducting Monte Carlo uncertainty analysis using 1,000 simulations.
- Performing computational validation and consistency checks.
- Comparing alternative model structures.
- Critically evaluating model assumptions and limitations.
- Organising a reproducible computational research workflow for GitHub.

PhD project statement

This project was undertaken as an independent computational public-health modelling exercise to develop practical understanding of mathematical modelling, Python-based simulation, uncertainty analysis and model evaluation. The project progressed from a deterministic SIR model to an SEIR extension and incorporated intervention analysis, sensitivity analysis, Monte Carlo uncertainty analysis and computational validation. The outputs are intended as methodological demonstrations rather than empirical forecasts of a real-world epidemic.